

Deployment Guide: Aegivex AI

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Document Type: Deployment Guide

Project: OKX.AI Genesis Hackathon

Prepared For: Development Team

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1. Overview

This document describes the deployment process for Aegivex AI. The objective is to deploy the application securely, reliably, and efficiently for production use while ensuring scalability and maintainability.

2. Deployment Architecture

Frontend → Vercel → **Backend API** → Railway / Render → **PostgreSQL Database** → OpenAI API → **Blockchain APIs**

3. Deployment Requirements

Required Services:

- GitHub
- Vercel
- Railway or Render
- PostgreSQL Database
- OpenAI API Key
- Domain Name (Optional)

4. Frontend Deployment

Technology: Next.js

Platform: Vercel

Deployment Steps:

1. Push project to GitHub.
2. Connect repository to Vercel.
3. Configure environment variables.

4. Build application.
5. Deploy automatically.

Environment Variables:

NEXT_PUBLIC_API_URL
NEXT_PUBLIC_APP_NAME

5. Backend Deployment

Technology: FastAPI

Platform: Railway or Render

Deployment Steps:

1. Push backend repository.
2. Install dependencies.
3. Configure environment variables.
4. Start FastAPI server.
5. Verify API endpoints.

Environment Variables:

DATABASE_URL
OPENAI_API_KEY
JWT_SECRET
JWT_EXPIRE
CORS_ORIGINS
LOG_LEVEL

6. Database Deployment

Database Engine: PostgreSQL

1. Create production database.
2. Run database migrations.
3. Verify table creation.
4. Create indexes.
5. Enable SSL connection.
6. Test database connectivity.

7. Environment Configuration

Scope	Variables
Frontend	NEXT_PUBLIC_API_URL, NEXT_PUBLIC_APP_NAME
Backend	DATABASE_URL, OPENAI_API_KEY, JWT_SECRET, JWT_EXPIRE, CORS_ORIGINS, LOG_LEVEL

Security Rule: Never expose secrets in frontend code.

8. Domain Configuration

Component	Domain URL
Frontend	https://app.aegivex.ai
Backend	https://api.aegivex.ai

Note: SSL certificates should be enabled for all domains.

9. Deployment Checklist

Category	Checklist Items
Frontend	Build successful No console errors Responsive UI verified

Category	Checklist Items
Backend	APIs working Authentication working AI integration working Database connected
Database	Migrations completed Indexes created Backups enabled

10. Monitoring

Metrics to Monitor: API uptime, Response time, Error rate, AI usage, Database performance, Server resources.

Recommended Tools: Sentry, UptimeRobot, Grafana, Prometheus.

11. Backup & Recovery

Database Backup: Daily incremental backup, Weekly full backup, Secure encrypted storage.

Recovery: Point-in-time recovery, Disaster recovery plan.

12. Security Checklist

Before production, verify the following:

- HTTPS enabled
- JWT authentication active
- Strong passwords enforced
- API rate limiting enabled
- CORS configured
- Input validation enabled
- SQL injection protection
- XSS protection
- Environment variables secured

13. CI/CD Pipeline

Recommended Workflow:

Developer Push → GitHub Repository → Automatic Build → Run Tests → Deploy Frontend → Deploy Backend → Health Check → Production Ready

14. Rollback Strategy

If deployment fails:

- Roll back to previous stable version.
- Restore previous database backup if required.
- Verify API functionality.
- Notify development team.
- Investigate deployment logs before redeployment.

15. Production Validation

After deployment, verify the following components:

- User Registration & Login
- AI Chat
- Wallet Scanner
- Token Analyzer
- Smart Contract Analyzer
- Website Scanner
- Transaction Explainer
- Dashboard & Scan History
- Notifications

16. Future Deployment Enhancements

Planned improvements:

- Kubernetes Deployment
- Docker Containers
- Auto Scaling
- CDN Integration
- Multi-Region Deployment
- AI Load Balancing
- Enterprise Infrastructure

17. Expected Outcome

The deployment process ensures that Aegivex AI is securely deployed, highly available,

scalable, and production-ready. The platform should provide reliable AI-powered Web3 security services with minimal downtime and a seamless user experience.